

Numerical Explorations of Two-Matrix Models with Various Potentials

Project Report



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Abstract

In this report we discuss the Monte Carlo simulation results of one-dimensional two-matrix models with various potentials. In addition to considering a harmonic-oscillator type matrix model, we also study gauged quantum mechanics matrix models with mass-square, harmonic oscillator, commutator, and quartic type potentials. All the models were simulated using hybrid Monte Carlo algorithm. The observables such as Polyakov loop, extent of space, and two-point correlators were computed.

Acknowledgements

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Last but not the least, I would like to thank my family for providing me with necessary moral and financial support.

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1 Introduction

Matrix models have become an interesting field in theoretical physics, especially string theory. In the community, there exists a model called the BFSS matrix model, which describes a one-dimensional theory with sixteen supercharges. In the large- N limit this model captures M-theory.

For simplicity, in this project, we will consider the Euclidean version of the bosonic BFSS model with two matrices, instead of nine. The action for this model can be obtained from dimensional reduction of the 3+1-dimensional Yang Mills theory with gauge group $SU(N)$ to 0+1-dimensions. After Wick rotation, we get the following form for the Euclidean action

$$S = \frac{N}{\lambda} \int_0^\beta dt \operatorname{Tr} \left\{ \sum_{i=1}^2 \frac{1}{2} (\mathcal{D}_t M^i)^2 - \frac{1}{4} \sum_{i,j=1}^2 [M^i, M^j]^2 \right\}, \quad (1)$$

where the matrices M^i are traceless Hermitian $N \times N$ matrices, the covariant derivative is

$$\mathcal{D}_t M^i = \partial_t M^i - i[A(t), M^i],$$

and $\lambda = g^2 N$ is the 't Hooft coupling. The action is invariant under $SU(N)$ gauge transformations

$$M^i(t) \rightarrow U(t) M^i(t) U^\dagger(t) \quad (2)$$

$$A(t) \rightarrow U(t) (A(t) + i\partial_t) U^\dagger(t). \quad (3)$$

In this project we will explore actions with various forms of the potentials, $V(M^i)$, in addition to the commutator type potential given in Eq. (1). We have the action

$$S_V = \frac{N}{\lambda} \int_0^\beta dt \operatorname{Tr} \left\{ \sum_{i=1}^2 \frac{1}{2} (\mathcal{D}_t M^i)^2 + \sum_{i=1}^2 V(M^i) \right\}. \quad (4)$$

The partition function is given as

$$Z = \int \mathcal{D}A \mathcal{D}M e^{-S_V}. \quad (5)$$

The expectation value of an observable O can be computed as

$$\langle O \rangle = \int \mathcal{D}A \mathcal{D}M O e^{-S_V}. \quad (6)$$

2 Two-Matrix Model on a Lattice

To discretize the model on a lattice we first break up the Euclidean time direction t in equally spaced pieces, as $t \in \{t = ja | j \in Z_{N_t}\}$, with $aN_t = \beta$. Further, we make the following analogies (in congruence with statistical mechanics and quantum field theory correspondence)

$$\int_0^\beta \rightarrow a \sum_{j=0}^{N_t}, \quad (7)$$

$$\partial_t M(t) \rightarrow \frac{M_{j+1} - M_j}{a}. \quad (8)$$

The covariant derivative is discretized as

$$\mathcal{D}_t M(t) \rightarrow \frac{U_{j,j+1} M_{j+1} U_{j,j+1}^\dagger - M_j}{a}. \quad (9)$$

The quantity of interest is $(\mathcal{D}_t M(t))^2$. Upon using $\text{Tr}(XY) = \text{Tr}(YX)$ and $\text{Tr}(X+Y) = \text{Tr}(X) + \text{Tr}(Y)$ the action is

$$S_V = \sum_{i=1}^2 \frac{Na}{\lambda} \sum_{n=0}^{N_t-1} \left\{ \frac{1}{2} \text{Tr} \left[\left(\frac{U_{n,n+1} M_{n+1}^i U_{n,n+1}^\dagger - M_n^i}{a} \right)^2 \right] + \text{Tr} \{ V(M_n^i) \} \right\}. \quad (10)$$

Expanding the kinetic term

$$\begin{aligned} \frac{1}{2} \text{Tr} \left\{ \left(\frac{U_{n,n+1} M_{n+1}^i U_{n,n+1}^\dagger - M_n^i}{a} \right)^2 \right\} &= \frac{1}{a^2} \text{Tr} \left\{ -M_n^i U_{n,n+1} M_{n+1}^i U_{n,n+1}^\dagger \right\} \\ &\quad + \frac{1}{a^2} \text{Tr} \{ (M_n^i)^2 \}. \end{aligned} \quad (11)$$

Absorbing the last term into the potential the action becomes

$$\begin{aligned} S_V &= \frac{N}{a\lambda} \sum_{i=1}^2 \sum_{n=0}^{N_t-1} \text{Tr} \left\{ -M_n^i U_{n,n+1} M_{n+1}^i U_{n,n+1}^\dagger \right\} \\ &\quad + \frac{N}{\lambda} \sum_{i=1}^2 \sum_{n=0}^{N_t-1} \text{Tr} \left\{ \frac{(M_n^i)^2}{a} + aV(M_n^i) \right\}. \end{aligned} \quad (12)$$

Consider the following set of non-local transformations

$$M_n^i \rightarrow \tilde{U}_{n-1}^\dagger M_n^i \tilde{U}_{n-1}, \quad (13)$$

where

$$\tilde{U}_{n-1} = \prod_{j=0}^{n-1} U_{j,j+1}. \quad (14)$$

In the above, n ranges from 1 to $N_t - 1$ and for $n = 0$ we map it to itself. This is to ensure that for all n , excluding 0, we have

$$\tilde{U}_{n-1} U_{n,n+1}^\dagger \tilde{U}_n^\dagger = \mathbf{1}. \quad (15)$$

The non-local transformations given in Eq. (13) satisfy the constraint Eq. (15), and simplifies the action with nearest neighbour interactions. Since periodic boundary conditions are imposed, the coupling to the U fields manifests at the $(T-1, 0)$ link.

We have the action

$$\begin{aligned} S_V = & \frac{N}{a\lambda} \sum_{i=1}^2 \left\{ \sum_{n=0}^{N_t-2} \text{Tr} \{ -M_n^i M_{n+1}^i \} + \text{Tr} \{ -M_{N_t-1} W M_0 W^\dagger \} \right\} \\ & + \frac{N}{\lambda} \sum_{i=1}^2 \sum_{n=0}^{N_t-1} \text{Tr} \left\{ \frac{(M_n^i)^2}{a} + aV(M_n^i) \right\}, \end{aligned} \quad (16)$$

where

$$W = \tilde{U}_{T-2} U_{T-1,0}. \quad (17)$$

Note that, by construction, W is unitary and hence we could as well diagonalize it. We have

$$W = V D V^\dagger, \quad (18)$$

where

$$D = \text{diag}\{e^{i\theta_j}\}, \quad (19)$$

with $j \in Z_N$. Further, performing a global transformation

$$M_n^i \rightarrow V M_n^i V^\dagger \quad (20)$$

we have the action

$$\begin{aligned} S_V(M, D) = & \sum_{i=1}^2 \frac{N}{a\lambda} \left(\sum_{n=0}^{N_t-2} \text{Tr} \{ -M_n^i M_{n+1}^i \} + \text{Tr} \{ -M_{N_t-1} D M_0 D^\dagger \} \right) \\ & + \frac{N}{\lambda} \sum_{n=0}^{N_t-1} \text{Tr} \left\{ \frac{(M_n^i)^2}{a} + aV(M_n^i) \right\}. \end{aligned} \quad (21)$$

The partition function changes in the following way, with the change in measure

$$\begin{aligned}
Z &= \int \mathcal{D}A \mathcal{D}M e^{-S_V} \\
&= \int \mathcal{D}\{U\} \mathcal{D}M e^{-S_V[M, \{U\}]} \\
&\propto \int \mathcal{W} \mathcal{D}M e^{-S_V[M, D]}.
\end{aligned} \tag{22}$$

In the above, $\int \mathcal{D}\{U\}$ is $\text{vol}(SU(N))^{T-1}$. Also, by a property of the Haar measure we can make sense of the proportionality in the last step.

Avoiding all the details, and assuming that the integration over $\mathcal{W}$ is correct for the purpose of computation, we have

$$Z \propto \int \mathcal{D}M e^{-S_V[M, D]} e^{S_{\text{FP}}[\theta]}, \tag{23}$$

$$S_{\text{FP}}[\theta] = -2 \sum_{j < k}^N \ln \left| \sin \left(\frac{\theta_j - \theta_k}{2} \right) \right|. \tag{24}$$

Since all observable are acquired through weighting, $e^{S_{\text{initial}} - S_{\text{final}}}$, we should face no problems in using the above Z for the simulation. More details regarding the lattice formulation is given in Ref. [2].

3 Implementation of HMC Algorithm

HMC algorithm, also known as the Hybrid Monte Carlo algorithm, is a robust way of performing Markov chain Monte Carlo simulations. Typically one may use the simpler version known as the Metropolis algorithm. However, it seems like for matrix models, the accuracy delivered and the computation time taken by HMC algorithm is promising. Comparisons have been neatly presented in Ref. [1] for matrix model harmonic oscillator. The main approach is to devise a Hamiltonian and subsequently arrive at the flow equations, which we integrate out, to reach the equilibrium field configurations. Following which, we use the crux of the Metropolis reject/accept approach and calculate observables.

The Hamiltonian for us is given by

$$H(P_m, P_j, M, \theta) = \sum_{n=0}^{N_t-1} \frac{1}{2} \text{Tr}\{(P_n^i)^2\} + \sum_{j=0}^{N-1} \frac{1}{2} \text{Tr}\{P_j^2\} + S[X, D(\theta)] + S_{\text{FP}}[\theta]. \tag{25}$$

The Hamilton's equations are

$$\dot{X}_n^i = P_n^i, \quad (26)$$

$$\dot{\theta}_j = P_j, \quad (27)$$

$$\dot{P}_n^i = -\frac{\partial H}{\partial X_n^i}, \quad (28)$$

$$\dot{P}_j = -\frac{\partial H}{\partial \theta_j}. \quad (29)$$

Listing out the last two equations for completeness, we have

$$-\frac{\partial H}{\partial X_0^i} = \frac{N}{a\lambda} (X_1^i - 2X_0^i + D^\dagger X_{N_t-1}^i D) + \frac{Na}{\lambda} \frac{\partial V}{\partial X_0^i}, \quad (30)$$

$$-\frac{\partial H}{\partial X_{N_t-1}^i} = \frac{N}{a\lambda} (X_{N_t-2}^i - 2X_{N_t-1}^i + DX_0^i D^\dagger) + \frac{Na}{\lambda} \frac{\partial V}{\partial X_{N_t-1}^i}, \quad (31)$$

$$-\frac{\partial H}{\partial X_n^i} = \frac{N}{a\lambda} (X_{n+1}^i - 2X_n^i + X_{n-1}^i) + \frac{Na}{\lambda} \frac{\partial V}{\partial X_n^i}, \quad (32)$$

$$-\frac{\partial H}{\partial \theta_j} = \frac{2N}{a\lambda} \sum_{k=1}^N \text{Re} (iX_{T-1,kj}^i X_{0,jk}^i e^{i(\theta_j - \theta_k)}) + \sum_{k=1, k \neq j}^N \cot \left(\frac{\theta_j - \theta_k}{2} \right). \quad (33)$$

Added to this, since the gauge group is $SU(N)$, we have the constraint $\sum_j \theta_j = 0$. To ensure that we do not have any runaway issues, we impose that M s and P s are traceless throughout the simulation. The basics of Metropolis algorithm are nicely explained in Ref. [1].

4 Observables of Interest

We will look at the following observables in our simulations.

4.1 Polyakov Loop

$$\langle |P| \rangle = \frac{1}{N} \left\langle \left| \mathcal{P}(e^{i \oint dt A}) \right| \right\rangle = \left\langle \left| \frac{\text{Tr}(D)}{N} \right| \right\rangle. \quad (34)$$

4.2 Extent of Space

$$\langle R^2 \rangle = \left\langle \frac{\lambda}{NN_t} \sum_{n=0}^{N_t-1} \text{Tr} \{ (M_n^i)^2 \} \right\rangle. \quad (35)$$

4.3 Correlation Functions

We will look at four types of correlators. They are

$$G_1(n) = \langle \text{Tr} \{ M_0^1 M_n^1 \} \rangle, \quad (36)$$

$$G_2(n) = \langle \text{Tr} \{ M_0^1 M_n^2 - M_0^2 M_n^1 \} \rangle, \quad (37)$$

$$G_3(n) = \langle \text{Tr} \{ M_0^1 M_n^2 \} \rangle, \quad (38)$$

$$G_4(n) = \langle \text{Tr} \{ M_0^1 M_n^2 + M_0^2 M_n^1 \} \rangle. \quad (39)$$

We note that all these correlators can be scaled by $1/N$. In the plots of the correlators we will show later, this scaling was removed. The motivation to consider these correlators comes from Ref. [3].

5 Two-Matrix Models with Various Potentials

We have considered four models.

The first one is the harmonic oscillator model without a gauge field. It has the action

$$\begin{aligned} S_{\text{HO}} = & \frac{N}{a\lambda} \sum_{i=1}^2 \left(\sum_{n=0}^{N_t-2} \text{Tr} \{ -M_n^i M_{n+1}^i \} + \text{Tr} \{ -M_{N_t-1} M_0 \} \right) \\ & + \frac{N}{\lambda} \sum_{i=1}^2 \sum_{n=0}^{N_t-1} \text{Tr} \left\{ \frac{(M_n^i)^2}{a} + a \frac{m^2 (M_n^i)^2}{2} \right\}. \end{aligned} \quad (40)$$

In the next three models, we collect the parts of the action that are common to all the potentials, and define a quantity $\mathcal{K}$. We have

$$\mathcal{K} \equiv \frac{N}{a\lambda} \sum_{i=1}^2 \left(\sum_{n=0}^{N_t-2} \text{Tr} \{ -M_n^i M_{n+1}^i \} + \text{Tr} \{ -M_{N_t-1} D M_0 D^\dagger \} \right). \quad (41)$$

We have mass-square potential model with the action

$$S_{\text{MSQ}} = \mathcal{K} + \frac{N}{\lambda} \sum_{i=1}^2 \sum_{n=0}^{N_t-1} \text{Tr} \left\{ \frac{(M_n^i)^2}{a} + a \frac{m^2 (M_n^i)^2}{2} \right\} + S_{\text{FP}}. \quad (42)$$

The commutator potential model with the action

$$S_{\text{COMM}} = \mathcal{K} + \frac{N}{\lambda} \sum_{i=1}^2 \sum_{n=0}^{N_t-1} \text{Tr} \left\{ \frac{(M_n^i)^2}{a} - a \frac{([M_n^i, M_n^j])^2}{4} \right\} + S_{\text{FP}}. \quad (43)$$

The quartic potential model with the action

$$\begin{aligned}
S_{\text{QUARTIC}} = & \mathcal{K} + \frac{N}{\lambda} \sum_{i=1}^2 \sum_{n=0}^{N_t-1} \text{Tr} \left\{ \frac{(M_n^i)^2}{a} + a \frac{m^2 (M_n^i)^2}{2} \right\} \\
& + \frac{a\kappa}{4} \text{Tr}\{(M_n^i)^2\}^2 + S_{FP}.
\end{aligned} \tag{44}$$

6 Simulation Details

We set $N = 8$, $N_t = 10$, $a = 0.\bar{3}$, and $\lambda = 1$ in the simulations. The lattice spacing value corresponds to a temperature $T = 0.3$. The step size for the Monte Carlo jumps are fine tuned for each model such that the acceptance rate is in the range $70\% \rightarrow 80\%$. All the variables and parameters are measured in units of the 't Hooft coupling λ_0 . By a suitable rescaling we can make all of them dimensionless.

We ran the simulations of all the models for 1000K sweeps. The simulation time for all sweeps roughly took 16 hours.

The Hamiltonian flow was solved for using leapfrog method. The number of leapfrogs was fixed at 10 for all simulations.

Once we have evaluated all the observables, we need to perform the autocorrelation algorithm to arrive at accurate averages and standard deviations. In total we had to find the autocorrelation lengths of 42 observables in each simulation. For brevity this report does not contain the autocorrelation plots for each observable. It was observed that the autocorrelation lengths for observables in models S_{HO} , S_{QUARTIC} and S_{MSQ} (with $m = 12$) were in the range $[10, 135]$. For models in S_{COMM} and S_{MSQ} (with $m = 0.276$) were in the range $[1000, 2000]$.

7 Simulation Results

7.1 S_{HO}

Our simulation results for this model are given in Figs. 1 and 2. The plots in Fig. 1 serve as good diagnostic plots to check if the simulation is running well. Fig. 1(a) describes the acceptance rate at every 100 sweeps. The red line indicates the mean, the legend shows the error. Fig. 1(b) describes $e^{\Delta H}$ at every 100 sweeps. The red line again indicates the mean. If the HMC simulation is working fine then the mean should be approximately 1. Fig. 1(c) describes nature of the R^2 in the simulation. The extent of space will increasingly grow if we don't demand for traceless matrices

during the simulation. In conclusion these preliminary results have conveyed to us that the code produces meaningful results and can be used for other potentials.

Fig. 2 has all the correlators plotted against lattice site. Fig. 2(a) is correct and matches with the refined plots seen in Refs. [1] and [2]. The other correlators need to be 0, since there is no spontaneous symmetry breaking in the model. We see that they roughly fluctuate about 0, and are in the range $[-4, 4] \times 10^{-4}$.

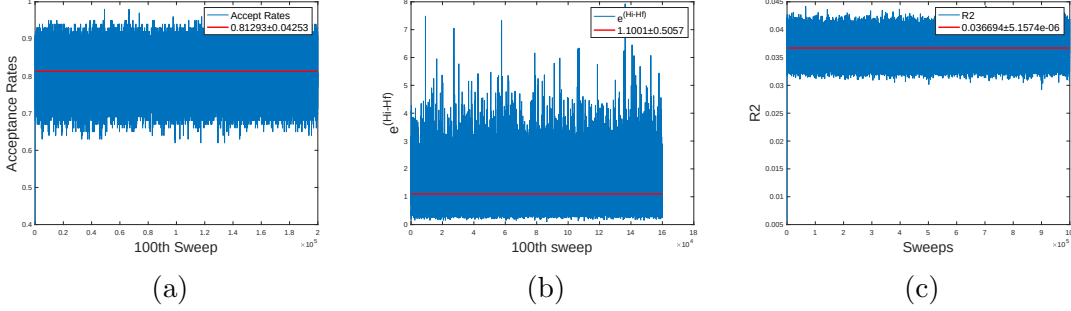


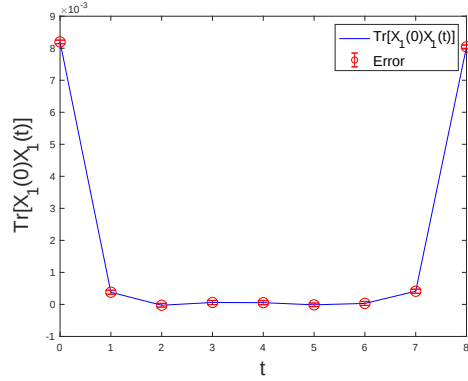
Figure 1: The model with the action S_{HO} . (a) Acceptance rate, (b) $e^{-\Delta H}$, and (c) Extent of Space.

7.2 S_{MSQ}

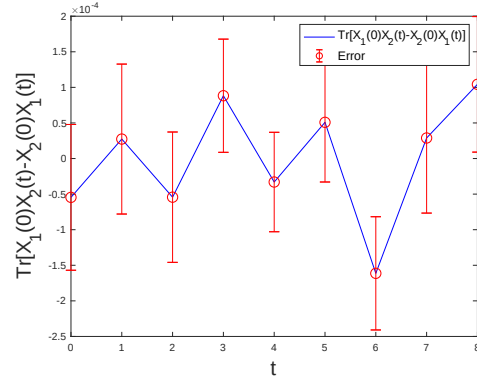
The results for the mass-square model is given in Figs. 3 and 4 for $m = 0.276$, and in Figs. 5 and 6 for $m = 12$. The reason to change the mass couplings was to see its influence in calculating the correlators. The Polyakov loop value indicates that we are in the confined phase for both the masses, at the same temperature. R_2 seems to be lower for the one with larger mass, which is a promising trend. The key feature here is that the $m = 12$ plots seem to take the correlators closer to 0.

7.3 S_{COMM}

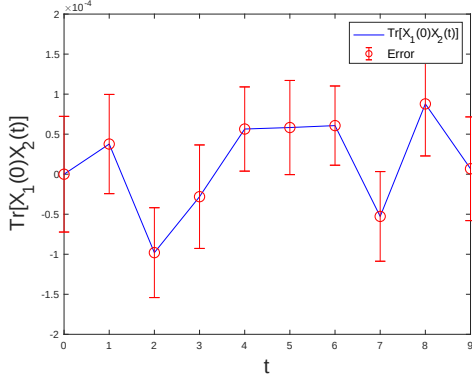
In Figs. 7 and 8 we show the results for the commutator model. The diagnostic plots are looking fine. Fig. 7(c) indicates that we are in the confined phase, with mean Polyakov loop value 0.116, which is roughly close to $0.125 = 1/N$ and matches with our expectation. All the plots given in Fig. 8 have been merely an exploration. It seems like G_1 is exponentially decaying. We also note that we do not see any periodicity. A comment to be written here is that the actions with gauge coupling have an asymmetry, so we should expect $n = 1$ and $n = N_t - 2$ to have similar



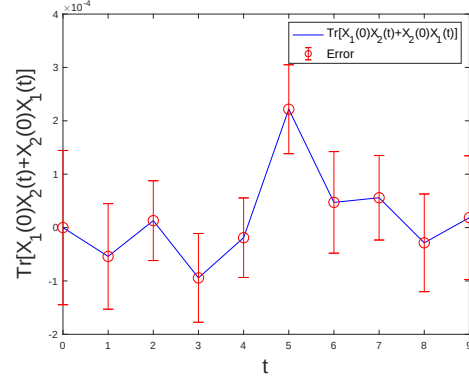
(a)



(b)



(c)



(d)

Figure 2: The model with the action S_{HO} . The correlators are shown in subfigures. (a) G_1 , (b) G_2 , (c) G_3 , and (d) G_4 .

values for the observable. One sees that this is clearly not present even in S_{HO} . In addition, since this model has a coupling between the scalars, we must expect non-zero correlators in the confined phase.

7.4 S_{QUARTIC}

In Figs. 9 and 10 we show the simulation results for the model with the quartic potential. The parameter values for these plots are $m = 12$ and $\kappa = 1$. We are in the weakly coupled regime. The correlator plots seem to fluctuate about 0.

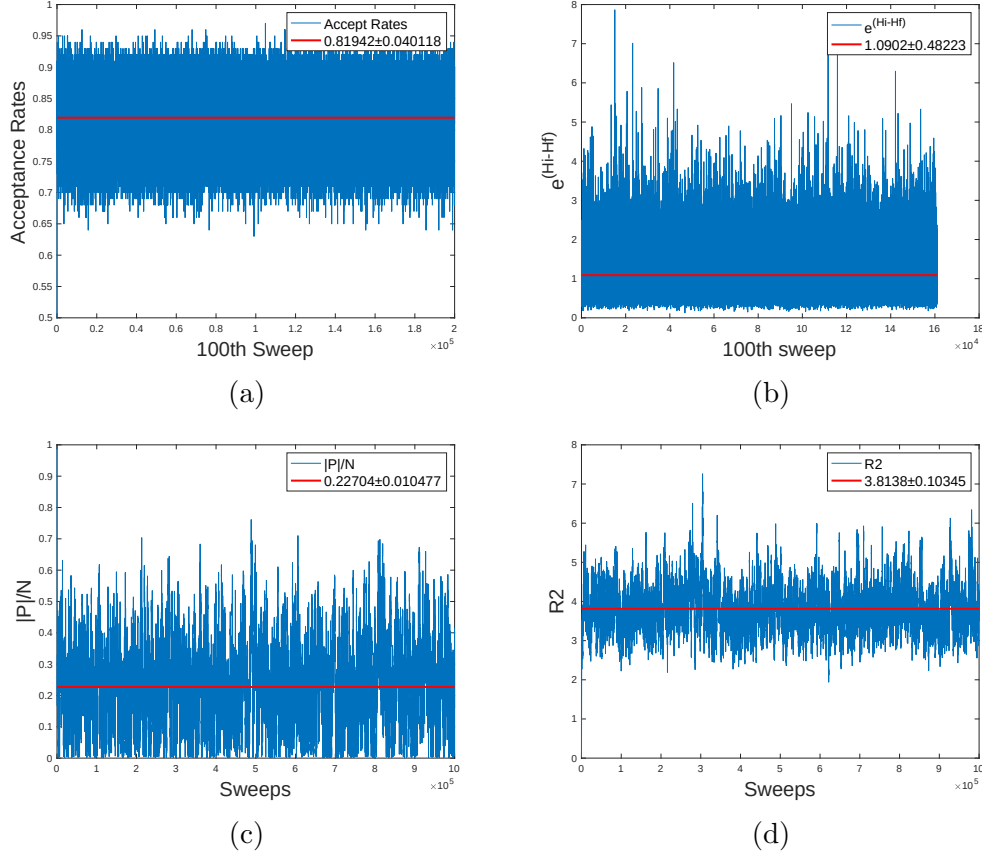


Figure 3: The model with the action S_{MSQ} and with $(m = 0.276)$. (a) Acceptance rate, (b) $e^{-\Delta H}$, (c) Polyakov loop, and (d) Extent of space.

8 Conclusions and Future Directions

The main motivation to calculate these observables was to find if there was a spontaneous symmetry breaking of Z_2 symmetry of the scalars at low temperatures. This would indicate a non-zero value for the correlators. It is clearly observed that the actions we have considered possess global symmetries of the kind Z_2 and $Z_2 \times Z_2$. In the case of S_{HO} , theoretically one expects the correlators to be 0, indicating no spontaneous symmetry breaking. This has been verified to some extent in our plots. Introducing a gauge symmetry (model S_{MSQ}), provided some non-triviality to the problem. The non-trivial Hilbert space makes one to revisit the question of a possi-

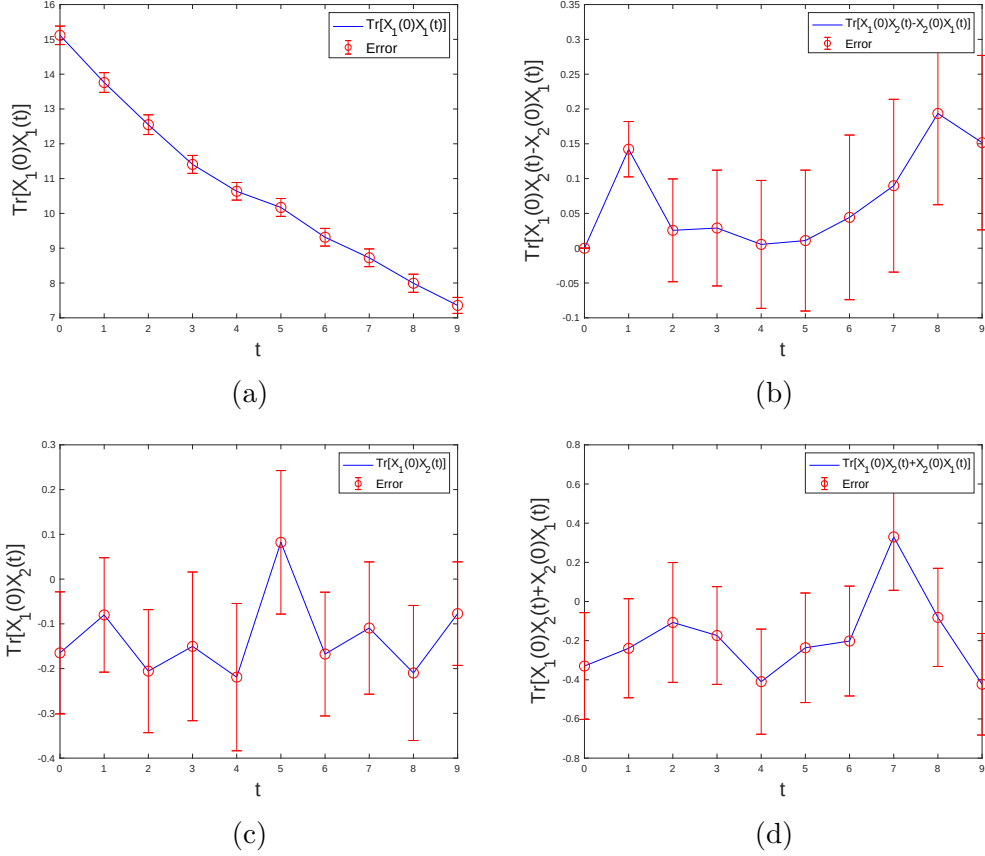


Figure 4: The model with action S_{MSQ} and with $(m = 0.276)$. The correlators are shown in the subplots. (a) G_1 , (b) G_2 , (c) G_3 , and (d) G_4 .

ble symmetry breaking¹.

In spirit of the original question, we have stayed staunch in the analysis of the correlators. We have shown that for a model with 8 colors and 10 sites the correlators seem to fluctuate very close to 0. We also observe that smaller mass couplings lead to slightly non-zero correlators. In the weak coupling for S_{QUARTIC} a similar trend prevails. At the moment we cannot make any strong claims or observations with our results.

Questions remain if all our models have a confined and deconfined phase. It is our claim that we are in the confined phase due to the nature of our Polyakov loop order parameter. Future directions will need to address this question by probing

¹These points have been encapsulated in Ref. [3]

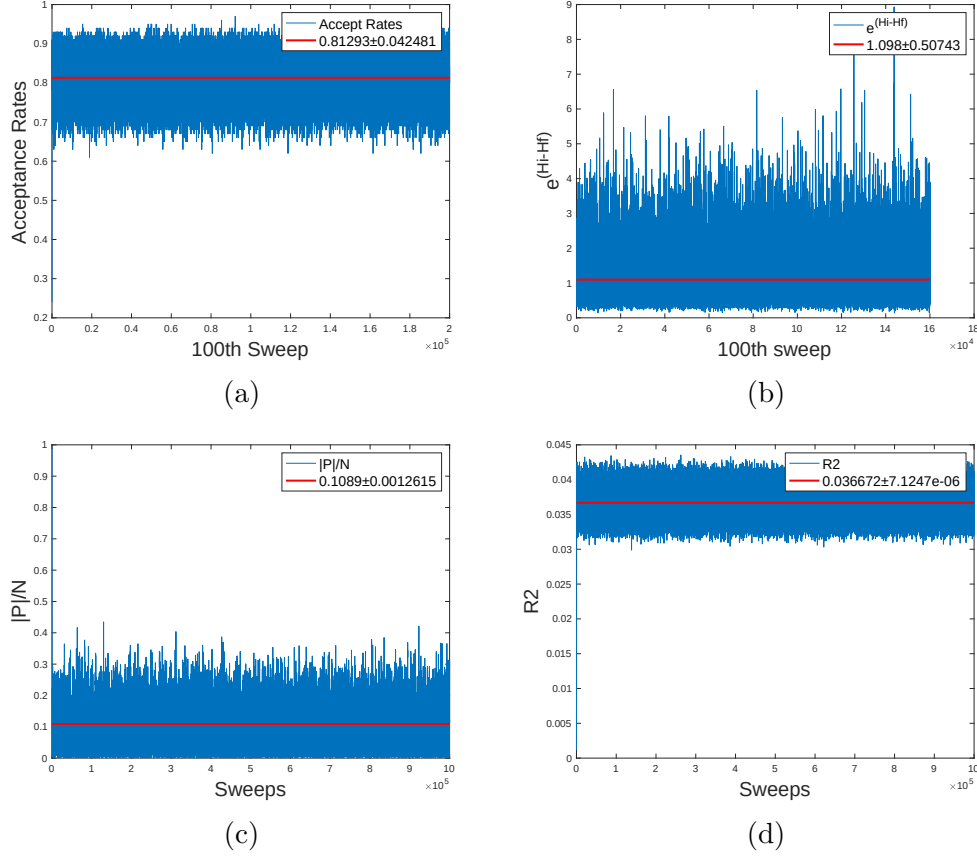


Figure 5: The model with action S_{MSQ} and with $(m = 12)$. (a) Acceptance rate, (b) $e^{-\Delta H}$, (c) Polyakov Loop, (d) Extent of Space.

more temperatures and hence significant work towards reducing the running time for codes. Another direction is to rerun the existing code for $N > 8$ and see the behaviour of correlators. We believe that for large number of colors at the current temperatures we are probing, the expectation is that the correlators are 0.

Questions regarding the periodic properties of the graphs remain. These will be immensely important in the context of using more scalars and also questioning the similarities between gauged and non-gauged models.

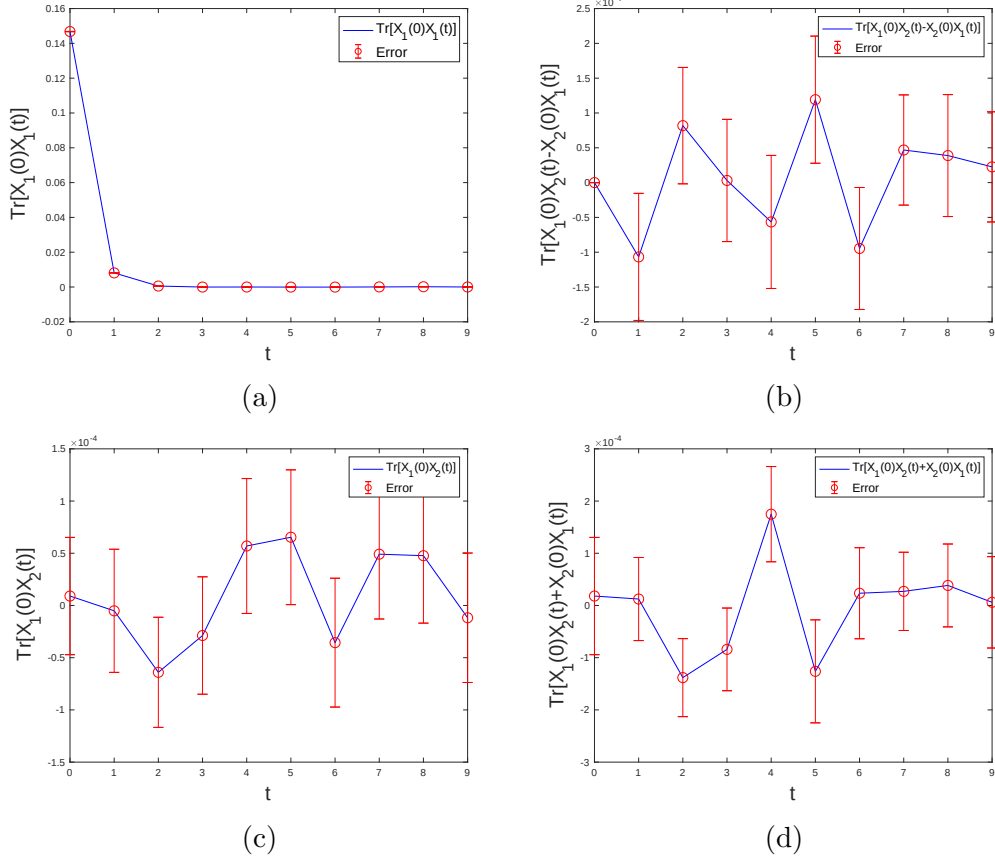


Figure 6: The model with action S_{MSQ} and with $(m = 12)$. The correlators are shown in the subplots. (a) G_1 , (b) G_2 , (c) G_3 , (d) G_4 .

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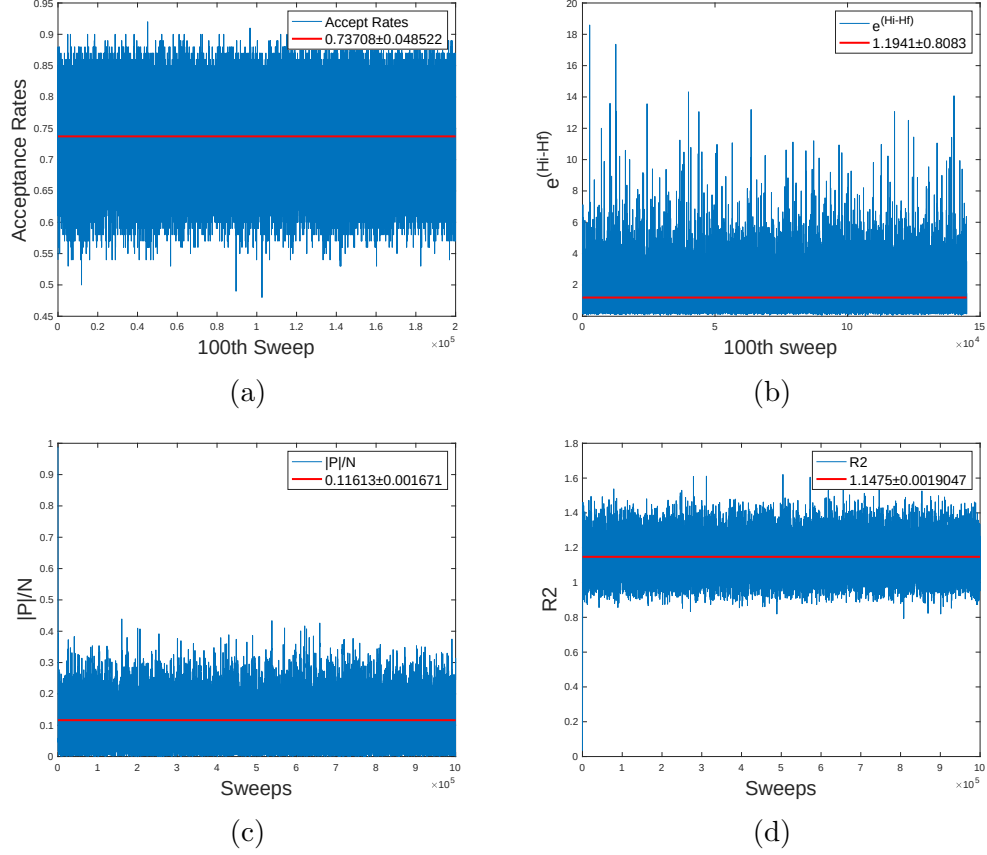
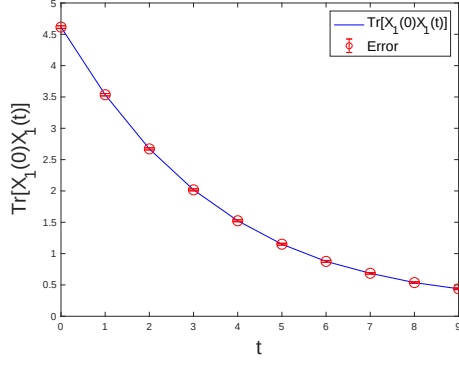
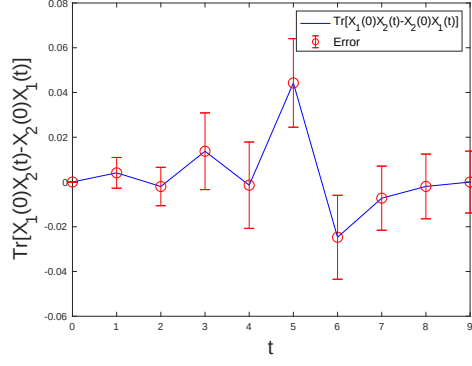


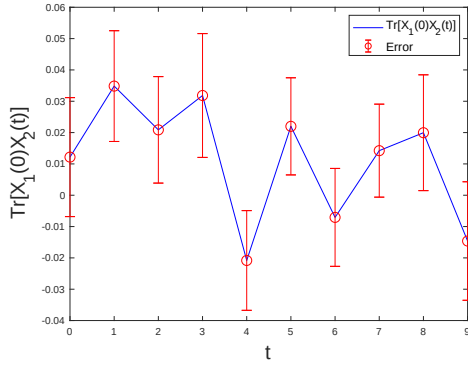
Figure 7: The model with commutator potential action, S_{COMM} . (a) Acceptance rate, (b) $e^{-\Delta H}$, (c) Polyakov Loop, AND (d) Extent of Space.



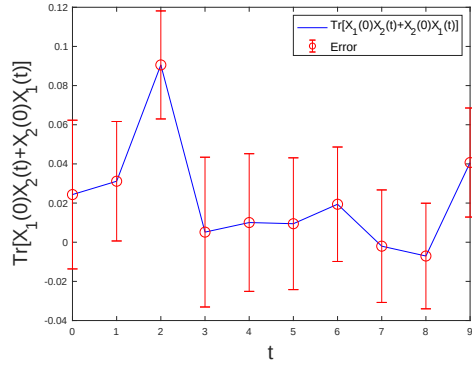
(a)



(b)



(c)



(d)

Figure 8: The model with commutator potential action, S_{COMM} . The correlators are shown in the subplots. (a) G_1 , (b) G_2 , (c) G_3 , and (d) G_4 .

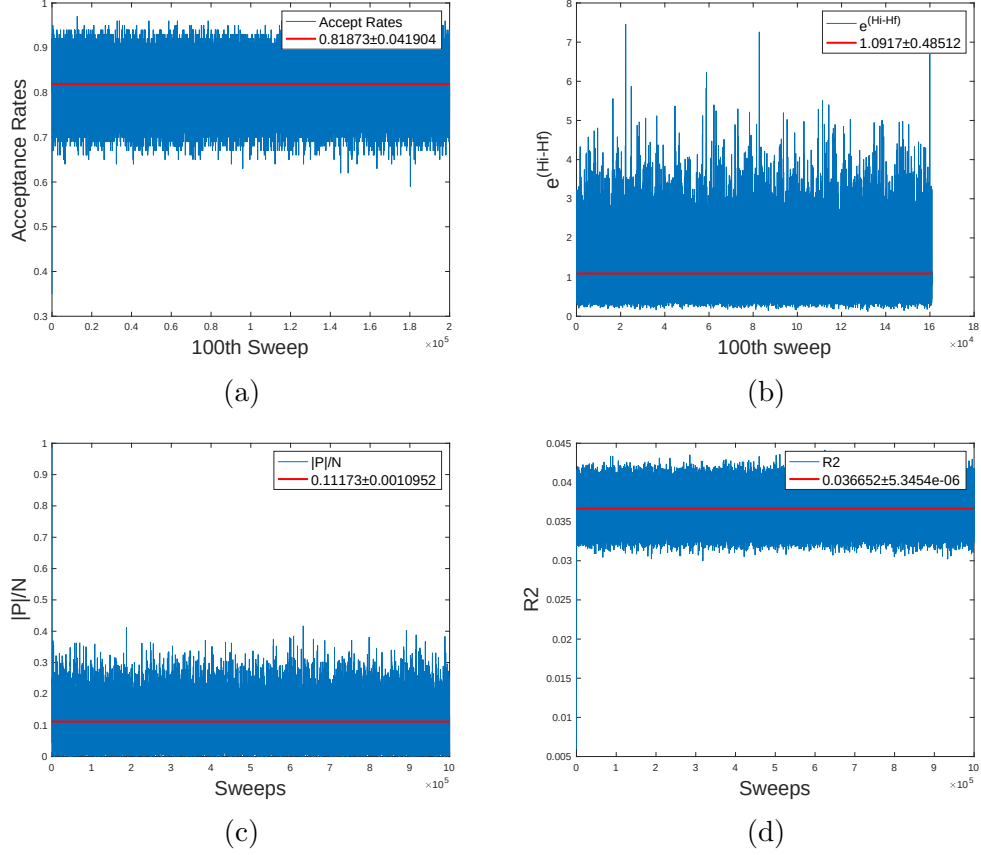


Figure 9: The model with quartic potential action S_{QUARTIC} with $m = 12$ and $\kappa = 1$. (a) Acceptance rate, (b) $e^{-\Delta H}$, (c) Polyakov loop, and (d) Extent of space.

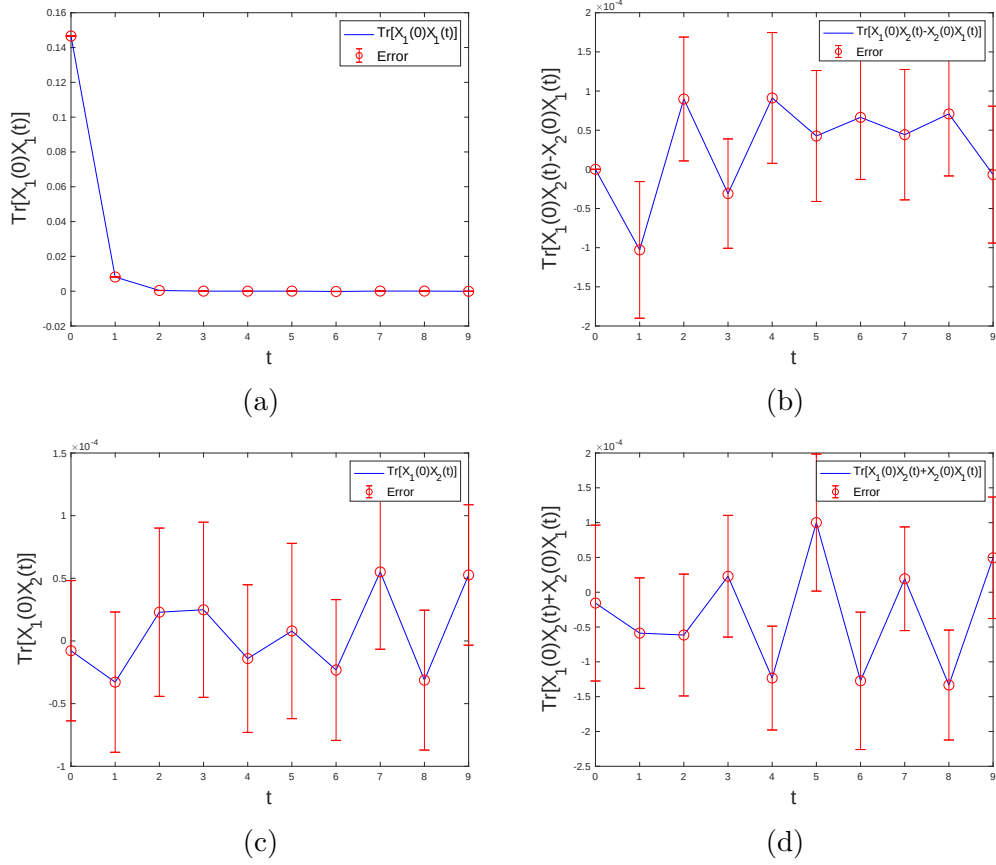


Figure 10: The model with quartic potential action S_{QUARTIC} with $m = 12$ and $\kappa = 1$. The correlators are shown in the subplots. (a) G_1 , (b) G_2 , (c) G_3 , and (d) G_4 .